The pending sign as a recursive operator: an anchor note on branching, growth modes, and their limits

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Abstract

This is an anchor note, not a results paper. It records what a focused exploration established about the pending-sign mechanism of the geometric companion — specifically, whether that mechanism is *recursive* (whether a resolved branch opens further branches) and whether its recursion carries physical content about how structure grows. Three findings are recorded, each with its epistemic status made explicit and each rechecked numerically: (i) the mechanism is recursive and its recursion is log-periodic with period $\ln 2$, intrinsically, because squaring a scale doubles its position on a logarithmic axis; (ii) the recursion exhibits two arithmetic modes — introducing a new prime ("innovation") and repeating an existing prime ("crystallization") — and the first pure-crystallization level coincides, by an independent route, with the first nested pending sign $I_{\mathrm{or}} > 0$ already identified in the octave structure; (iii) the proton sits as the base case of the recursion, its branches 3 and 5 around the root 2^2 being the simplest instance of the same octave law. The note is equally explicit about what these findings do *not* establish: they are properties of the arithmetic of the Mersenne spectrum, verified within the model, and they do not by themselves demonstrate that physical crystals, genes, or discharge structures grow by this mechanism. The selection rule that would make the recursion a model of real branching — "new information is a new multiple" — is recorded as the current conceptual frontier, resolved in arithmetic form but not yet connected to a measurable physical prediction. The purpose of anchoring is to fix these points so a later iteration can build from them without re-deriving.

1 Why this note exists

The geometric companion introduced orientation as a *pending sign*: the undetermined branch of a square root, fixed only upon projection, with the proton as its concrete instance (the root $4 = \sqrt{16} = 2^2$ leaving a pending ± 1 that projects the branches 3 and 5). A natural question was left open there: is that mechanism *recursive*? Does a resolved branch itself carry a further pending sign, so that the operation can be applied to its own output? The question matters because recursion is the difference between a single differentiation and a growth engine: crystals, biological hierarchies, and discharge (lightning) structures all grow by applying a branching rule to its own output, level after level. If the pending sign is recursive, it is a candidate operator for such growth; if not, it is a single act of differentiation.

This note records the outcome of probing that question. It is written as an anchor: a fixed reference fixing what was found, with each claim labeled by status, so that a later iteration can proceed from here. Following the program's discipline, the labels are: *Finding* (a checked mathematical fact about the model), *Observation* (a numerical fact about a particular case), *Conceptual* (an interpretation consistent with the model but not forced by it), and *Open* (not established).

2 Finding 1: the recursion exists and is log-periodic of period $\ln 2$

Finding 1 (Recursive self-similarity). *The pending-sign mechanism is recursive: each value n, taken as a centre, has n^2 as its root-square, and the pending ± 1 projects branches $n^2 - 1$ and $n^2 + 1$, with*

$$n^2 - 1 = (n - 1)(n + 1),$$

a difference of squares — the same octave law $M_{2k} = M_k Q_k$ that governs the transfer ledger. The operation therefore reproduces its own form at every level. Because each level maps $n \mapsto n^2$, and $\ln(n^2) = 2 \ln n$, the recursion doubles *position on a logarithmic axis: it is log-periodic with period $\ln 2$.*

This was checked on an explicit lineage $2 \mapsto 3 \mapsto 8 \mapsto 63 \mapsto 3968 \mapsto \cdots$ (taking the $n^2 - 1$ value as the next scale): the successive $\log_2$ gaps are 0.585, 1.415, 2.977, 5.977, 11.954, whose ratios 2.42, 2.10, 2.008, 2.0001 converge to 2. Each squaring doubles the logarithmic gap, the signature of exact log-periodic self-similarity.

Status and significance. This is a *Finding*: it is intrinsic, following from the squaring operation that is part of the mechanism, not from any external choice. Its significance is that it *ties the pending-sign mechanism to the octave structure of the companion*: the resonance band's period-$\ln 2$ self-similarity and the branching tree's period-$\ln 2$ self-similarity are the same ladder, because both are the multiplicative-by-2 structure on the logarithmic axis. The snowflake-like branching and the $k \to 2k$ resonance signature are, at this level, one object.

What it does not establish. A tree that opens *all* branches at every level is not yet a model of a physical structure, which is *selective* — a real snowflake does not branch everywhere, it branches where local conditions permit. The unrestricted recursion, checked separately, reproduces the central binomial coefficients $(1, 2, 6, 20, 70, \dots)$, which is a property of the chosen recursion rule (stepping to $n \pm 1$), not of physics. Recognizing this guarded against attributing an artifact of the rule to the model. Selection is treated in Finding 2 and Section 5.

3 Finding 2: two growth modes, and an independent coincidence

Seeking the selection rule — what decides which branches open — led to the criterion, in the program's own terms, that *new information is a new multiple*: a level contributes genuinely new structure when it introduces arithmetic not already present. Made precise on the Mersenne spectrum, "new information" reads as a new *prime* entering the factorization of $M_\epsilon = 2^\epsilon - 1$.

Finding 2 (Two arithmetic growth modes). *Across depths ϵ, the Mersenne spectrum separates two modes:*

- Innovation — *M_ϵ introduces a prime not seen at any lower depth. These are the levels $\epsilon \in \{2, 3, 4, 5, 7, 8, 9, \dots\}$ (most levels).*
- Crystallization — *M_ϵ contains a repeated prime ($I_{\text{or}}(\epsilon) > 0$), reusing prime structure already present. These are the levels $\epsilon \in \{6, 12, 18, 20, 21, 24, \dots\}$.*

The two modes are not exclusive at every level (a level may both repeat a prime and introduce another, e.g. $\epsilon = 12$ repeats 3 and introduces 13), but the first pure-crystallization level*, introducing no new prime at all, is $\epsilon = 6$: $M_6 = 63 = 3^2 \cdot 7$, whose primes 3 and 7 both appeared earlier (3 at $\epsilon = 2$, 7 at $\epsilon = 3$).*

Observation 1 (Independent coincidence at $\epsilon = 6$)**.** *The first level that introduces* no new prime *(Finding 2) and the first level with a* nested pending sign $I_{\text{or}} > 0$ *(from the companion's octave structure) are the same level, $\epsilon = 6$. These were reached by two independent routes — one asking "where does no new prime information enter," the other asking "where does a prime first repeat" — and they coincide because a repeated prime is precisely the reuse of existing information. Two paths, one point.*

Status and significance. Finding 2 is a *Finding* (checked arithmetic); Observation 1 is an *Observation* of convergence between two descriptions already in the program. The *Conceptual* reading — that innovation and crystallization correspond to the two modes by which nature builds structure, variation (a mutating gene, new information) and iteration (a copying lattice, reused information) — is consistent with the mechanism and is recorded as such, not as a demonstrated correspondence. Its value is that the pending-sign ontology contains, intrinsically, a distinction between building-by-novelty and building-by-repetition, and that this distinction lands on the same levels the octave structure already flagged.

4 Finding 3: the proton as the base case

Finding 3 (Proton as base case of the recursion)**.** *The proton instance of the geometric companion is the base case of the recursion of Finding 1. The root $4 = 2^2$, with pending ± 1, projects the branches $3 = 2^2 - 1$ and $5 = 2^2 + 1$, whose product $3 \times 5 = 15 = M_4 = 2^4 - 1$ is the difference of squares $(2^2 - 1)(2^2 + 1)$ — the octave law at $k = 2$. The proton has $I_{\text{or}} = 0$: its branches are distinct primes, a completed differentiation (a branch split), not a nested pending sign.*

Status. *Finding* (checked). Its significance for this note is structural: it places the proton not as an isolated numerical coincidence but as the simplest instance — the first branching — of the same recursive operator whose deeper levels are the crystallization levels of Finding 2. The proton is where the tree begins.

5 The conceptual frontier, and its honest limit

The exploration reached a definite frontier, and the anchor's job is to mark it precisely.

What is resolved. The selection rule sought throughout — what inclines the pending sign, what decides which branches crystallize — has an arithmetic form: *new information is a new multiple (a new prime)*, and its absence (a repeated prime) is crystallization. This resolves, in the model's own arithmetic, the question of what distinguishes a growth step that innovates from one that repeats. The pending sign is genuine uncertainty; the branch that resolves is the one that constitutes a distinguishable scale; "distinguishable" is made precise as "new multiple." In the program's language: the information exists before the structure (both branches are latent in the pending sign), and structure crystallizes it (projection resolves the sign, and a new multiple — new information — appears). The nodes are seeds; crystallization levels are where existing information re-expresses rather than innovates.

What is open. **[Open]** This is an arithmetic correspondence within the model. It does *not* establish that physical crystals, genes, or lightning grow by this mechanism. Two structures sharing the property "self-similar growth with repetition" need not have one explaining the other; the coincidence may be generic. The step from "the Mersenne spectrum distinguishes innovation from crystallization" to "this is how real structure grows" is not authorized by the evidence in hand. Concretely, three things remain open:

- a *physical* prediction: does the innovation/crystallization distinction imply anything measurable about a real growing structure (a spectral feature, a scale ratio, a branching statistic), beyond the internal arithmetic?
- the *selection dynamics*: in a real medium the branch chosen depends on local conditions (field, temperature, binding); the model's rule is currently global arithmetic, with no coupling to a local environment. What plays the role of the local condition?
- the *status of* $\epsilon = 4$: the proton base case still rests on $\epsilon = 4$, which remains structurally motivated, not derived (as recorded in the companion).

Why anchor here. The exploration did what the program's method intends: it took a mechanism whose concept was unresolved (is the pending sign recursive, and does its recursion mean anything?) and resolved the arithmetic part while marking the physical part as open. The recursion is real and log-periodic; the two growth modes are real and coincide with the octave structure; the proton is the base case; and the leap to physical crystallization is explicitly *not* taken. Anchoring fixes these so the next iteration — most naturally, the search for a measurable consequence of the innovation/crystallization split, or the coupling of the selection rule to a local environment — can begin from a checked foundation rather than re-derive it.

6 Reproducibility: the Mersenne factorization table and the meaning of I_{or}

Two elements underlying Findings 2 and 3 are made explicit here so they can be verified without re-derivation: the factorization table that separates the two growth modes, and the exact reason I_{or} counts nested pending signs.

6.1 The factorization table

Table 1 lists $M_\epsilon = 2^\epsilon - 1$, its prime factorization, the count $I_{\text{or}}(\epsilon) = \sum_i(\alpha_i - 1)$, and the two growth-mode flags, for $\epsilon = 1$ to 24. The table is generated by a short self-contained script (standard library only; reproduced in the note's repository as `mersenne_spectrum.py`), so every entry is checkable. The separation is exact: a level is *innovation* if its factorization contains a prime absent at all lower depths, and *crystallization* if it contains a repeated prime ($I_{\text{or}} > 0$). The two are not exclusive — most crystallization levels also innovate — and only $\epsilon = 6$ is pure crystallization (repeats a prime, introduces none), which is why it is the first level with no new prime information and simultaneously the first with a nested pending sign.

6.2 Why I_{or} counts nested pending signs

The identification of $I_{\text{or}}(\epsilon) = \sum_i(\alpha_i - 1)$ with the number of nested pending signs is not an analogy; it follows from the pending-sign mechanism applied to a repeated prime. A pending sign is the undetermined branch of a square root: a magnitude is fixed, the sign is not, until

ϵ	M_ϵ	factorization	I_{or}	innov.	cryst.	new primes
1	1	1	0	–	–	(none)
2	3	3	0	yes	no	3
3	7	7	0	yes	no	7
4	15	$3 \cdot 5$	0	yes	no	5
5	31	31	0	yes	no	31
6	63	$3^2 \cdot 7$	1	no	**yes**	(none)
7	127	127	0	yes	no	127
8	255	$3 \cdot 5 \cdot 17$	0	yes	no	17
9	511	$7 \cdot 73$	0	yes	no	73
10	1023	$3 \cdot 11 \cdot 31$	0	yes	no	11
11	2047	$23 \cdot 89$	0	yes	no	$23, 89$
12	4095	$3^2 \cdot 5 \cdot 7 \cdot 13$	1	yes	**yes**	13
13	8191	8191	0	yes	no	8191
14	16383	$3 \cdot 43 \cdot 127$	0	yes	no	43
15	32767	$7 \cdot 31 \cdot 151$	0	yes	no	151
16	65535	$3 \cdot 5 \cdot 17 \cdot 257$	0	yes	no	257
17	131071	131071	0	yes	no	131071
18	262143	$3^3 \cdot 7 \cdot 19 \cdot 73$	2	yes	**yes**	19
19	524287	524287	0	yes	no	524287
20	1048575	$3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41$	1	yes	**yes**	41
21	2097151	$7^2 \cdot 127 \cdot 337$	1	yes	**yes**	337
22	4194303	$3 \cdot 23 \cdot 89 \cdot 683$	0	yes	no	683
23	8388607	$47 \cdot 178481$	0	yes	no	47, 178481
24	16777215	$3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$	1	yes	**yes**	241

Table 1: The Mersenne spectrum with growth modes. Crystallization levels ($I_{\text{or}} > 0$): $\{6, 12, 18, 20, 21, 24\}$. Pure-crystallization (repeat, no new prime): only $\epsilon = 6$. The count I_{or} peaks at $\epsilon = 18$ (3^3 gives two nested pending signs).

projection resolves it. When a prime p appears with multiplicity α in M_ϵ, the factor p^α is α nested copies of the same structure. Their relative orientations are the pending signs. Fixing one copy as the reference — the "-1" that poses the difference, since a global inversion of all copies is physically equivalent (as in the generative algebra's σ) — leaves the orientations of the remaining $\alpha - 1$ copies undetermined relative to it. Each of those is one pending sign. Summing over distinct primes, which are independent, gives

$$I_{\text{or}}(\epsilon) = \sum_i (\alpha_i - 1),$$

exactly the count in the table. The square-free case ($\alpha_i = 1$ for all i) gives $I_{\text{or}} = 0$: distinct primes are a completed differentiation (a branch split, like the proton's $3, 5$), with no residual pending sign. A repeated prime ($\alpha_i \geq 2$) is the first place a sign is left nested and unresolved. This is why "first repeated prime" and "first nested pending sign" are the same event, and why the count is $\alpha - 1$ rather than α: one copy is always the reference against which the others are differences.

7 Anchor summary

Statement	Status
Pending sign is recursive: $n \mapsto n^2 \mapsto (n^2-1,\, n^2+1)$	Finding (checked)
Recursion is log-periodic, period $\ln 2$ (from squaring = doubling in ln)	Finding (checked, intrinsic)
Tree and octave band share one $\ln 2$ ladder	Finding (checked)
Unrestricted tree reproduces central binomials	Observation (artifact of the open-all rule, not physics)
Two growth modes: innovation (new prime) vs crystallization (repeated prime)	Finding (checked; full table §6)
$I_{\text{or}} = \sum(\alpha_i - 1)$ counts nested pending signs	Derived (§6), not analogy
First pure-crystallization level = first $I_{\text{or}} > 0$ level $= \epsilon = 6$	Observation (convergence, two routes)
Innovation/crystallization $\leftrightarrow$ variation/iteration in nature	Conceptual (consistent, not demonstrated)
Proton is the base case: branches $3, 5$ around 2^2, product M_4	Finding (checked)
"New information = new multiple" as the selection rule	Resolved in arithmetic form; table & code in §6
Physical crystals/genes/lightning grow by this mechanism	Open (not established)
Measurable physical consequence of the two modes	Open
Coupling of the selection rule to a local environment	Open
$\epsilon = 4$ for the proton base case	Open (structurally motivated, not derived)

This note anchors the pending-sign recursion. Its solid content is arithmetic and internal: a real, log-periodic recursive operator with two growth modes, tied to the octave structure, with the proton as its base case. Its frontier is physical: whether this arithmetic is how structure grows in the world remains open, and is the natural subject of the next iteration.

References

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